Harbor Grid Resilience Review

HGR-2026-17 - synthetic long-form evaluation report

Executive summary

This review examines the Harbor West electricity resilience programme across 18 substations during the April to September 2026 observation period. The work combined incident logs, maintenance records, operator interviews and metering data so that conclusions did not depend on a single operational source. The programme was designed to reduce avoidable interruptions while preserving the conservative isolation rules used around cranes, ferry charging equipment and refrigerated container yards.

Headline finding

The evidence indicates that reliability improved, although the effect was uneven between districts. Automated feeder checks reduced repeated dispatches in East Basin, while South Pier continued to experience restoration delays during night shifts. The report therefore recommends targeted staffing and relay calibration instead of a network-wide change to protection thresholds.

How to read this report

Figures and tables use validated operational data. Narrative observations provide context but do not override a logged event. Where a metric changed between the baseline and the post-upgrade period, both pages must be read together. Nearby percentages and training examples are intentionally retained because they reflect the ambiguity found in real reports and create useful retrieval negatives.

Scope boundary

The review covers operational resilience, energy use and governance. It does not contain customer contact details, payment credentials, private addresses or access passwords. Questions seeking those facts should be treated as unsupported rather than answered from background knowledge.

Recommendation sequence

Management should first complete the label programme and night-shift staffing trial, then observe a second matched period before changing protection settings. This sequence preserves a stable technical baseline while addressing the operational weaknesses most strongly supported by the evidence. The review does not recommend purchasing a new control platform, because software availability was not a material contributor to the measured interruptions.

1. Programme context and method

Long-form prose, references and exact identifiers

Operating environment

Harbor West is a mixed industrial precinct whose daily demand shifts with vessel arrivals, cold-storage loading and ferry charging. Unlike a residential feeder, its peak can move by several hours from one day to the next. Operators therefore evaluate both absolute demand and the sequence of equipment starts before changing a switching plan. The programme identifier used in maintenance systems is HBR-OPS-41; HBR-OPS-14 is an older training scenario and is not part of this review.

Evidence collection

Analysts reconciled supervisory control records with field tickets and weekly planning notes. A timestamp was accepted only when two independent logs agreed within one minute. Meter readings were normalized for temperature and scheduled vessel activity. Interview statements were used to explain decisions, but every quantitative claim in the findings is tied to a table, incident record or approved calculation.

Review design

The baseline covered eight complete weeks before the relay and scheduling changes. The comparison period used eight matched weeks after commissioning and excluded a planned port-wide shutdown. This design does not establish causality on its own, but it reduces obvious seasonal and workload differences. Sensitivity checks retained storm days and then removed them; the direction of the principal findings remained unchanged.

Terminology

An interruption begins when service falls outside the validated operating envelope, not when the first warning appears. Restoration is recorded when stable supply has been maintained for five minutes. A dispatch is a physical field response, whereas a remote intervention is performed by the control room. These distinctions matter when matching questions to evidence.

Quality controls

Two analysts independently reviewed unusual events and reconciled disagreements against the original log sequence. Calculations were reproduced from exported interval data, and every manually corrected record retained its earlier value in an audit column. Site names were normalized only after the original label had been preserved. These controls make the fixture useful for questions that distinguish an approved fact from a plausible nearby value.

2. Incident chronology

Timeline narrative with nearby but non-equivalent times

Initial warning

At 04:11 on 18 June, the monitoring service raised a temperature warning for the South Pier feeder. The warning did not itself interrupt supply and did not authorize isolation. The duty controller compared the reading with the parallel sensor, checked the vessel schedule and requested a field confirmation because recent maintenance had changed the sensor housing.

Protective action

At 04:17 the controller isolated feeder SP-3 after the parallel sensor confirmed a sustained rise. This is the authoritative isolation time. A radio note created at 04:19 describes the isolation retrospectively, while a maintenance ticket opened at 04:23 records the technician response. Those later timestamps are workflow events and must not be substituted for the operational action.

Restoration

The inspection found a loose termination rather than a failed relay. After torque verification and a thermal scan, the feeder returned to service at 04:46. Stable supply was confirmed at 04:51 under the report's five-minute rule. The incident therefore contributed 34 interruption minutes to the operational series even though the elapsed time from first warning to stable confirmation was 40 minutes.

Lessons

The review found that the controller followed the escalation procedure and that the early warning created useful preparation time. It also found that the radio and ticket systems described the same event with different timestamps. Retrieval tests should distinguish the requested event type before selecting a time.

Related events

Two events from the same month were excluded from this chronology. A 03:52 warning at North Quay cleared without isolation, and a 05:08 ticket at West Yard concerned planned testing. They share equipment language with the South Pier incident but do not supply its action time, feeder identifier or restoration duration. Their inclusion here is intentional: realistic reports often discuss adjacent events that are topically similar but evidentially irrelevant.

3. Reliability profile by site

Native-text table embedded in explanatory prose

Interpretation

Table 1 summarizes validated incidents after the upgrade. Peak demand is descriptive and is not an alert threshold. The event count includes both remote and field-restored interruptions, while median restore time uses the operational definition introduced in Section 1.

Site	Peak MW	Events	Median restore	Owner
North Quay	7.4	6	11 min	Grid Control
East Basin	5.9	4	7 min	Field Services
South Pier	8.1	8	14 min	Grid Control
West Yard	6.7	5	9 min	Asset Care
Harbor Central	9.3	3	6 min	Network Ops
Dry Dock	4.8	7	13 min	Field Services
Container Gate	7.9	5	10 min	Asset Care
Ferry Terminal	5.2	2	5 min	Network Ops

Reading the table

South Pier has the highest event count and the longest median restore time among the listed sites. Harbor Central has the greatest peak demand but only three validated events, so peak demand alone does not explain incident frequency.

4. Governance and field observations

Two-column policy discussion with role boundaries

Decision rights

Routine switching plans are approved by the Network Operations Lead. Emergency isolation remains with the duty controller because waiting for a committee could extend a hazardous condition. Changes to protection settings follow a different path: the Reliability Review Board examines the engineering case, and the Asset Director provides final authorization after independent safety review.

Meeting evidence

The board met on 7 July with six voting members present. The quorum rule requires five voting members, so the decision was valid. Two observers from port operations attended but did not vote. Minutes refer to an earlier workshop attended by eight people; that attendance number is not the board quorum and should not be used to answer governance questions.

Field feedback

Technicians reported that clearer device labels shortened handovers and reduced calls to the control room. They also noted that wet-weather access at South Pier remained slow. The feedback supports the staffing recommendation but does not establish the amount of reliability improvement, which comes from the incident analysis.

Control recommendation

The report recommends retaining current trip thresholds, completing label replacement by 30 November 2026 and assigning an additional night-shift technician on high-traffic days. It explicitly rejects a proposal to raise thresholds simply to reduce alert volume.

Accountability trail

Progress is reported monthly to the board secretary, but completion evidence is accepted by the Asset Assurance Manager. Procurement can confirm delivery of replacement labels; it cannot declare the operational action complete. This distinction prevents a delivery date from being mistaken for the approved installation deadline.

5. Baseline energy analysis

First half of a cross-page comparison

Measurement boundary

Energy analysis covers auxiliary cooling and control equipment, not crane propulsion or tenant loads. Readings were collected at fifteen-minute intervals and aggregated only after clock drift and missing intervals were resolved. Three short gaps were filled using adjacent validated intervals; the imputed share was below one percent of the baseline total.

Baseline result

During the matched pre-upgrade weeks, auxiliary systems used an average of 18.4 MWh per operating day. The median was 18.1 MWh and the highest single day reached 22.7 MWh during a heat event. The daily average, 18.4 MWh, is the baseline required for the programme comparison.

Nearby measures

A separate facilities dashboard reports 17.6 MWh for administrative buildings. That number uses a different meter boundary and must not be combined with the harbor auxiliary series. The training worksheet also contains an illustrative 20 percent saving, but it is not an observed result.

Uncertainty

Meter calibration uncertainty is estimated at plus or minus 0.3 MWh per day. Because the post-upgrade change exceeds this band, the review treats the direction of change as credible while avoiding claims of precision beyond one decimal place.

Baseline distribution

Weekday averages were more stable than weekend averages because vessel arrivals concentrated auxiliary demand into predictable windows. No single site accounted for more than one third of the baseline series. Removing the highest-demand day reduced the average slightly but did not change the value selected for the matched comparison, which was calculated under the frozen analysis protocol.

6. Post-upgrade energy outcome

Second half of a cross-page comparison

Observed outcome

Across the matched post-upgrade weeks, auxiliary systems used an average of 15.1 MWh per operating day. The reduction was concentrated in overnight cooling cycles and did not coincide with an increase in temperature alarms. This page must be combined with the 18.4 MWh baseline on the preceding page to calculate the absolute programme change.

Derived comparison

The difference between the two validated daily averages is 3.3 MWh per operating day, equivalent to a reduction of approximately 17.9 percent relative to baseline. The report uses the unrounded values for internal calculations and publishes the one-decimal figures so readers can reproduce the stated comparison.

Alternative explanations

Weather normalization accounts for part of the raw reduction, while changed vessel activity accounts for little of it. The review cannot exclude all behavioral effects because technicians knew the equipment was being monitored. Even so, sensitivity tests produced reductions between 15.8 and 18.6 percent.

Decision

The programme should continue for another six months with the same meter boundary. Expanding the scope before that point would make the time series difficult to compare and would weaken the audit trail.

Monitoring plan

The next review will report both the matched daily average and the full-period total so that operational growth is visible without changing the comparison definition. Analysts will also track temperature alarms, manual overrides and deferred maintenance. A reduction in energy use will not be considered successful if it coincides with weaker cooling performance or delayed safety action.

7. Interruption results by district

Raster-only figure with surrounding narrative

Figure guidance

Figure 2 reports average interruption minutes after the upgrade for four districts. The chart values exist only in the raster image; this paragraph deliberately does not repeat them. A correct visual answer must inspect the bar length or printed value and keep the district-to-value relationship intact.

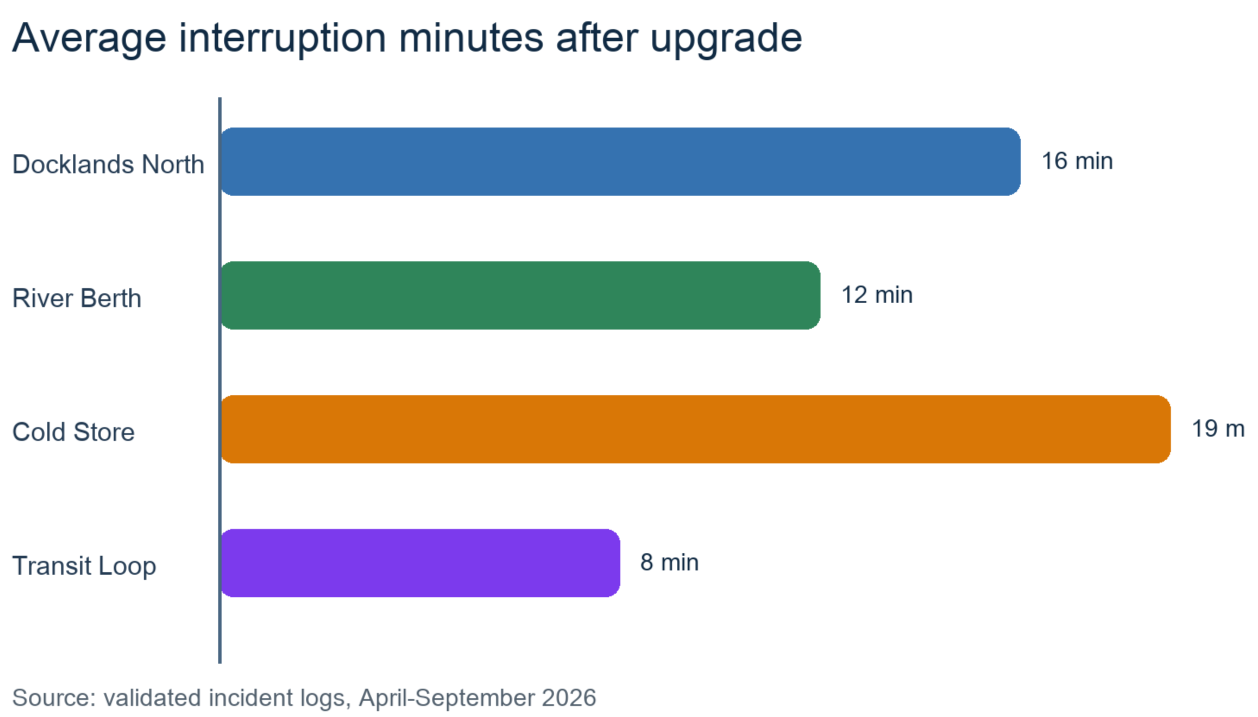

Figure 2. Average interruption minutes after upgrade. Values are raster-only.

8. 中文现场复核记录

中文长段落、近义概念与精确事实

复核范围

蓝港现场复核覆盖东港池与南码头两个区域，重点检查夜班交接、设备标签和告警确认流程。复核人员将控制室日志、现场工单和班组记录逐项核对，只有时间、设备编号和处置动作能够相互印证时，才把该记录纳入正式统计。访谈内容用于解释操作背景，但不能单独作为数值结论。

关键要求

中文作业补充规程的编号是 CN-HARBOR-62。规程要求当冷却回路连续三次超过 74 摄氏度时启动人工复核；单次短暂告警只记录观察，不立即改变运行状态。培训材料中的 47 摄氏度是演示值，不是触发阈值。

交接发现

抽查显示，多数班组能够说明告警确认与设备隔离的区别，但两个夜班记录把首次告警时间误写成处置时间。项目组因此要求交接表分别保留告警、确认、隔离和恢复四个字段，避免后续报告把不同事件混为一谈。

改进期限

设备标签更换应在 2026 年 11 月 30 日前完成，责任团队为现场服务组。这个期限与英文治理章节一致；会议纪要中出现的 10 月 30 日只是供应商提供标签样稿的日期，不能作为最终完成期限。

后续验证

完成标签更换后，质量人员将抽取白班和夜班记录，检查四类时间字段是否齐全，并确认现场编号与控制室编号能够对应。抽查通过不代表可以修改保护阈值；任何阈值调整仍需按照英文治理章节规定的审批路径执行。

9. Version notes, limitations and negative evidence

Conflicting values are labelled rather than silently removed

Version statement

This document is version 1.0, approved on 12 October 2026. A draft circulated in August proposed a 72 degree cooling threshold, but that proposal was rejected. The approved Chinese supplement sets the threshold at 74 degrees after three consecutive exceedances. Systems answering from the approved version must not revive the draft value.

Known limitations

The observation period is too short to estimate rare major failures, and no conclusion is drawn about equipment outside the defined meter boundary. Ferry demand changed during two weeks, although the matched analysis reduces its influence. Results should not be generalized to residential networks without further evidence.

Unsupported requests

The report contains no employee telephone numbers, customer account numbers, passwords, private addresses or banking information. It also does not name an insurance provider. A related topic or plausible-looking identifier is not evidence that one of these facts exists.

Archival note

All source logs used for this synthetic fixture are represented by deterministic generated content. The document is designed for repeatable evaluation and contains no production data or real user identifiers.

Citation guidance

A supported answer should cite the page containing the requested operational fact, not merely this limitations page. When a question compares baseline and post-upgrade performance, both analysis pages are required even though this section summarizes the document's status. Draft values may be mentioned only when the answer clearly identifies them as rejected and cites the approved replacement.